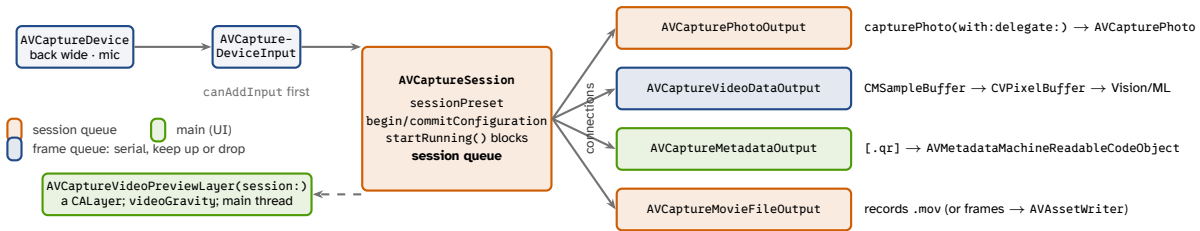


**In one line:** AVCaptureSession is a **graph**: devices → inputs → session → outputs, configured and started on a **private serial queue** (startRunning() blocks), frames delivered on **another** serial queue, UI on main. AVAudioSession is **policy, not playback**: category + mode + options tell iOS whether you mix, duck, obey the silent switch, record, and keep playing in the background.



## Capture

- Permission:** `NSCameraUsageDescription` (+ `...Microphone...`) — missing key = **crash** on access. `AVCaptureDevice.authorizationStatus(for:)` → `requestAccess(for:)` (completion off main; async overload); `.denied/.restricted` → `Settings`. Mic alone: `AVAudioApplication.requestRecordPermission` (iOS 17).
- Configure** between `beginConfiguration()` and `commitConfiguration()` (atomic camera/preset swap); `canAdd...` before every `add...`. Focus/exposure/torch: `lockForConfiguration()` → `set` → `unlock` (check `is...Supported`).
- Frames:** `captureOutput(_:didOutput:from:)` at frame rate; `alwaysDiscardsLateVideoFrames` (default true) drops rather than queues — drops arrive in `didDrop`.
- Photo:** a **new** `AVCapturePhotoSettings` per shot; `photoQualityPrioritization` `.speed/.balanced/.quality`. Saving is the **Photos** framework's job (`PHPHotoLibrary`, add-only usage key).
- Scanning without the graph:** `VisionKit` `DataScannerViewController` (iOS 16) — text + barcodes, live UI.
- Rotation** (iOS 17): `videoRotationAngle` + `AVCaptureDevice.RotationCoordinator` replace `videoOrientation`.
- Interruptions:** `AVCaptureSessionWasInterrupted/...InterruptionEnded`; reasons e.g. `videoDeviceNotAvailableInBackground`, `...WithMultipleForegroundApps` (iPad multitasking), `...InUseByAnotherClient`. Runtime error → restart on the session queue. Hot device (`ProcessInfo.thermalState`) → lower preset/fps.

## Example — configure off main

```
func configure() { // runs on sessionQueue
    session.beginConfiguration()
    defer { session.commitConfiguration() }
    guard let cam = AVCaptureDevice.default(.builtinWideAngleCamera,
        for: .video, position: .back),
        let input = try? AVCaptureDeviceInput(device: cam),
        session.canAddInput(input) else { return }
    session.addInput(input)
    let frames = AVCaptureVideoDataOutput()
    frames.setSampleBufferDelegate(self, queue: videoQueue) // serial
    if session.canAddOutput(frames) { session.addOutput(frames) }
    let meta = AVCaptureMetadataOutput(); session.addOutput(meta)
    meta.setMetadataObjectDelegate(self, queue: .main)
    meta.metadataObjectTypes = [.qr] // AFTER addOutput
}
sessionQueue.async { configure(); session.startRunning() }
```

## Audio session — category = defaults

| Category       | silent sw. | mixes           | BG * | for              |
|----------------|------------|-----------------|------|------------------|
| .ambient       | silenced   | yes             | no   | game/UI sound    |
| .soloAmbient   | silenced   | no              | no   | <b>default</b>   |
| .playback      | plays      | no <sup>†</sup> | yes  | media, podcasts  |
| .record        | —          | no              | yes  | recorder         |
| .playAndRecord | plays      | no <sup>†</sup> | yes  | VoIP, video call |

\* also needs `UIBackgroundModes.audio`. <sup>†</sup> unless `.mixWithOthers` / `.duckOthers`.

.ambient/.soloAmbient also stop on screen lock.

- `setCategory(_:mode:options:)` → `setActive(true)` (fails during a call); end with `.notifyOthersOnDeactivation`.
- Modes** tune DSP/routing: `.moviePlayback`, `.spokenAudio`, `.voiceChat` (echo cancel), `.videoChat`, `.measurement`.  
**Options:** `.duckOthers`, `.mixWithOthers`, `.defaultToSpeaker` (else `playAndRecord` uses the *earpiece*), `.allowBluetoothA2DP`.
- Interruption:** `.began` — already stopped, update UI; `.ended` + `.shouldResume` → reactivate, resume. An `.ended` is not guaranteed.
- Route change** `.oldDeviceUnavailable` (headphones out) → **pause**. `mediaServicesWereReset` → rebuild all audio objects.
- Player:** `AVAudioPlayer` = local file, simple · `AVPlayer` = streams/HLS, video · `AVAudioEngine` = real-time node graph (mixer, effects, `installTap`); restart it on `AVAudioEngineConfigurationChange`.
- Lock screen = `MPNowPlayingInfoCenter` + `MPRemoteCommandCenter`; a `.mixWithOthers` app can't be `Now Playing`.

## Traps

- ! `startRunning()` on main = UI freeze; config from two threads = races. One serial session queue owns the session.
- ! `metadataObjectTypes` before `addOutput` (or an unsupported type) **raises an exception** — not a silent no-op.
- ! Keeping `CMSampleBuffers` past the callback starves the capture pool → dropped frames. Copy what you need.
- ! “No sound on silent” = default `.soloAmbient`; use `.playback`. `AVAudioPlayer` in a local `let` is deallocated and stops.

## Remember

**Three queues, one graph · category + mode + options = policy · pause on **began** and on **headphones out**.**

Related: concurrency (serial queues, main hop) · background-execution (audio mode) · rendering-pipeline (pixel buffers, image decode) · app-hardening-privacy (permissions) · state-restoration-multitasking (camera in iPad multitasking)